SMART ECG-BASED STRESS AND HEALTH ALERT SYSTEM

1. ABSTRACT

The Smart ECG-Based Stress and Health Alert System is designed to continuously monitor a user's heart activity using the AD8232 ECG module. This system detects stress or abnormal cardiac conditions in real-time by analyzing ECG signals and heart rate variability. The processed data triggers alerts through a GSM module (SIM800L), sending SMS notifications to caretakers or emergency contacts. The project focuses on developing a compact, affordable, and efficient system for preventive health monitoring.

2. INTRODUCTION

Heart health monitoring is a cornerstone of preventive healthcare. Electrocardiography (ECG) records the heart's electrical activity, identifying stress, arrhythmias, and abnormalities. While clinical-grade machines are accurate, they are often costly and require professional operation. This project utilizes the AD8232 analog front-end for efficient signal conditioning, integrated with an Arduino Uno for real-time processing. The system addresses the need for a low-cost, portable solution that identifies stress conditions and alerts users or caregivers instantly before severe complications occur.

3. OBJECTIVES

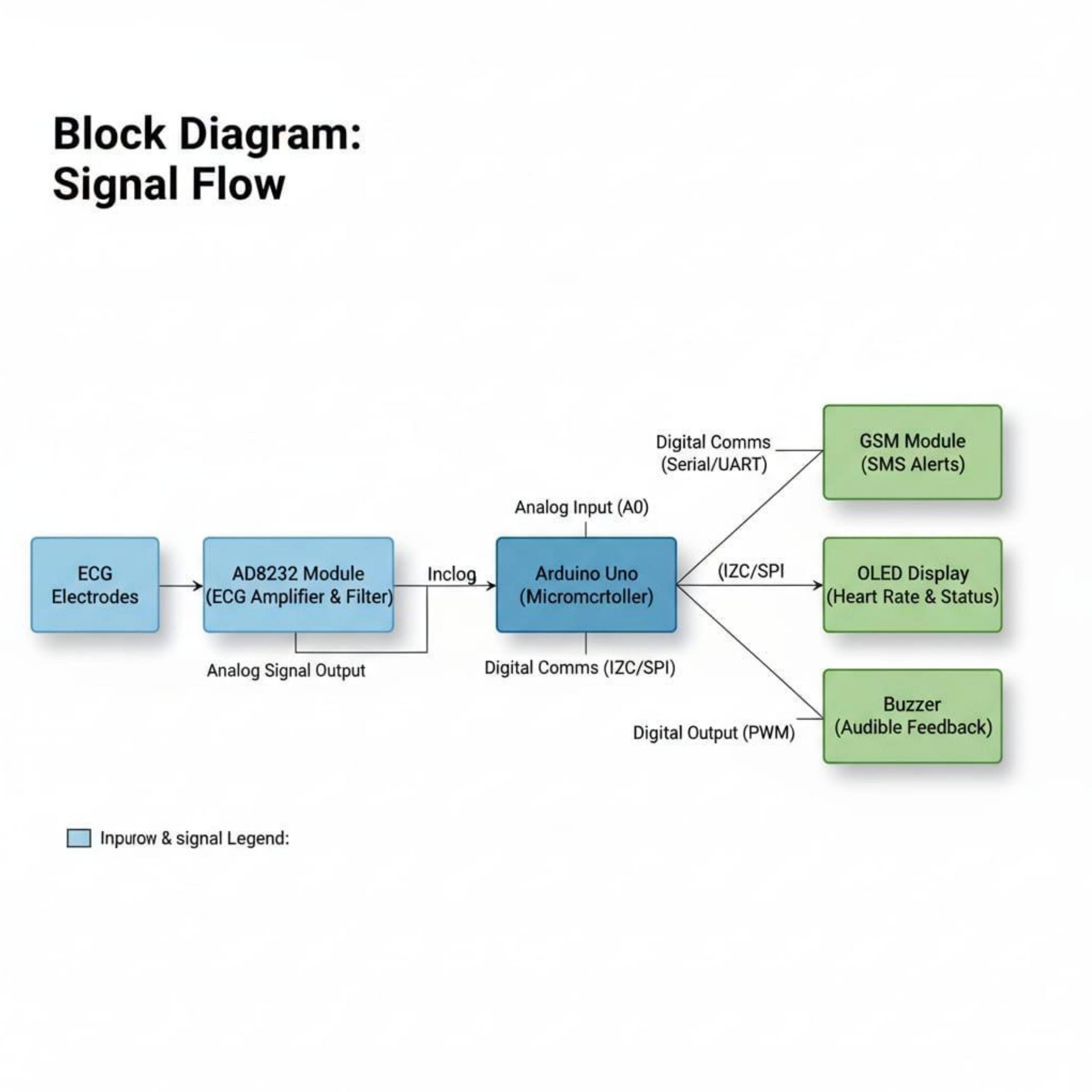
- Acquire ECG signals using the AD8232 analog front-end module.
- Process and analyze signals to determine Heart Rate (BPM) and stress levels.
- Design an alert system capable of notifying caretakers in case of irregularities using GSM.
- Display real-time data including BPM and ECG waveforms on an OLED screen.
- Develop an affordable and portable system for health monitoring applications.

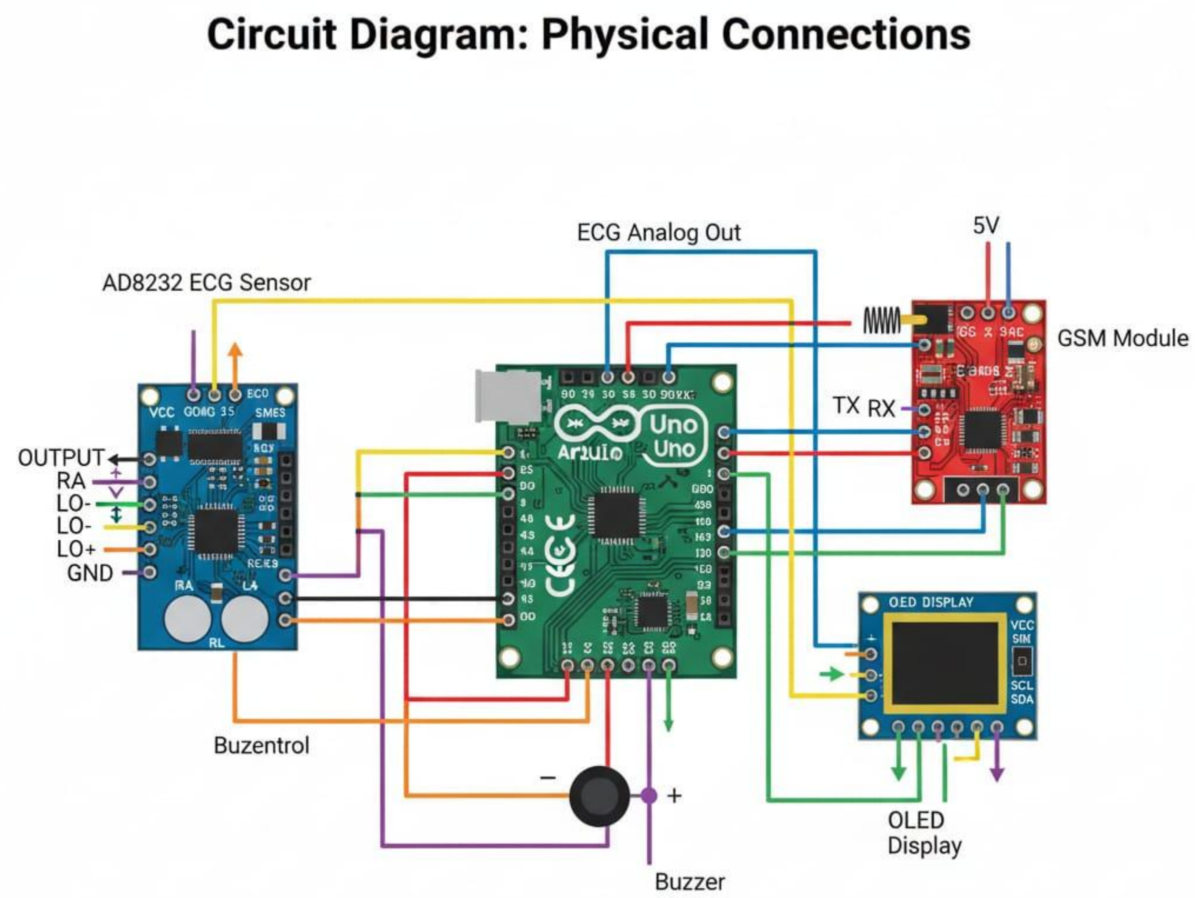
4. LITERATURE REVIEW

Research indicates that physiological stress directly influences Heart Rate Variability (HRV). Elevated stress levels result in increased BPM and irregular waveform patterns.

- Existing Systems: Hospital ECG machines are accurate but expensive, and many wearable devices lack robust real-time alert mechanisms.
- Limitations: Traditional equipment lacks continuous monitoring and instant alert systems.
- Opportunity: Integrating low-cost modules like the AD8232 with microcontrollers and GSM bridges the gap between clinical tools and personal health monitoring.

5. SYSTEM ARCHITECTURE & HARDWARE DESIGN





The architecture involves signal acquisition, filtering, amplification, processing, and alert generation.

Hardware Components

- Arduino Uno: The core controller that processes analog data and manages alert logic.
- AD8232 ECG Module: An analog front end that amplifies and filters biopotential signals.
- Electrodes: Placed on the RA, LA, and RL positions to capture heart signals.
- OLED Display: Shows BPM, stress status, and waveform patterns in real-time.
- GSM Module (SIM800L): Sends SMS alerts when anomalies or high stress are detected.

- LED Indicators: Provide visual classification (Green, Yellow, Red) of stress levels.
- Buzzer: Provides an audible signal for abnormal heart rates or stress.
- Power Supply: Powered via USB or a 9V battery for system portability.

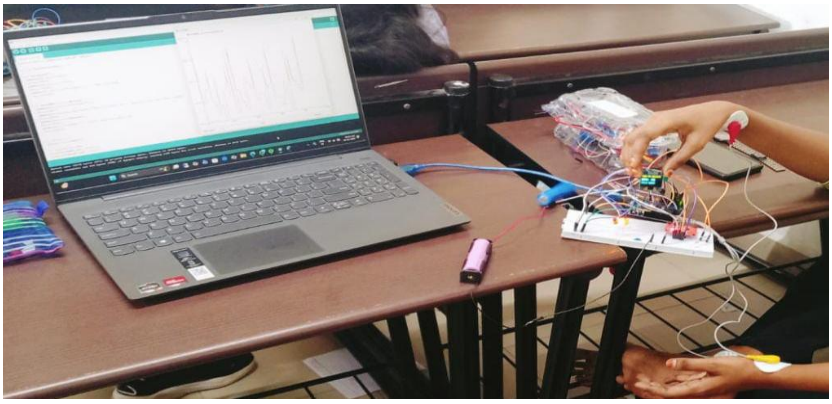
6. SOFTWARE IMPLEMENTATION

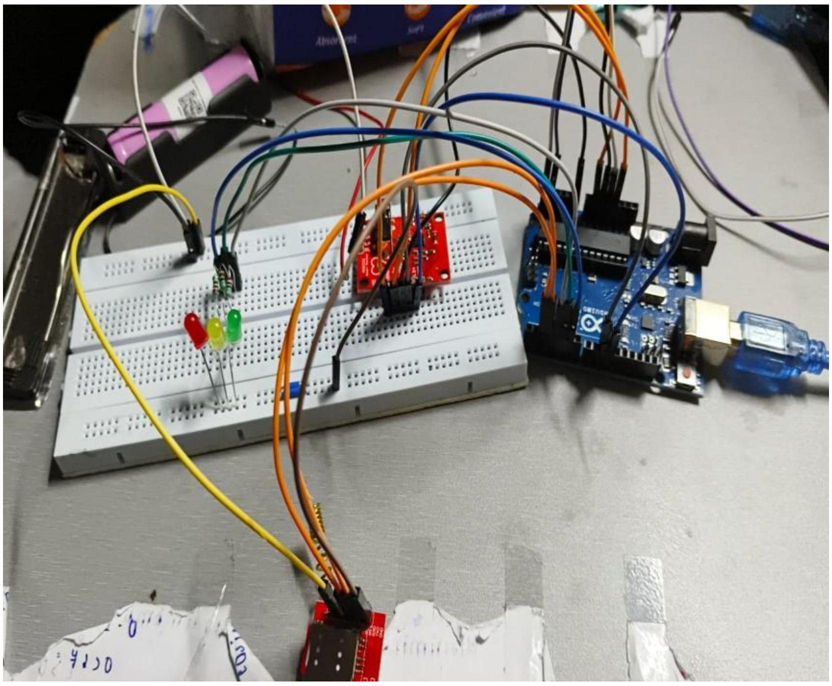
The system is programmed in Embedded C using the Arduino IDE.

Working Logic

1. Initialize: Set up the ECG module, OLED, GSM, and buzzer pins.
2. Read: Continuously acquire analog ECG data from pin A0.
3. Process: Apply threshold detection to identify R-peaks and calculate BPM.
4. Classify: Compare BPM with preset thresholds to determine the user's state (Normal, Mild Stress, or High Stress).
5. Alert: If BPM is >110 or <50, activate the buzzer and send an SMS alert via GSM.
6. Display: Update the OLED screen with the heart rate and detected condition.

7. TESTING AND RESULTS





- Performance: The system successfully captured ECG signals through the AD8232 module, producing noise-filtered waveforms.
- Observations: Under normal conditions, BPM remained between 70–85. During stressful conditions, BPM increased beyond 100, validating the system’s sensitivity.
- Accuracy: The response time was <2 seconds with high reliability under stable conditions.
- Alert Verification: The GSM module transmitted instant SMS alerts to registered contacts during abnormal readings.

8. APPLICATIONS & SUSTAINABILITY

- Applications: Personal health tracking, hospital/clinic monitoring, and stress management for students or professionals.
- Sustainable Development Goals (SDGs):
  - SDG 3: Promotes good health and well-being through early detection.
  - SDG 9: Encourages innovation in affordable healthcare technology.
  - SDG 11: Supports smart healthcare for safer communities.

9. CHALLENGES & FUTURE SCOPE

Challenges

- Movement affects accuracy, and noise can interfere with the ECG signal.
- GSM module power issues and limited battery life for portable use.

Future Scope

- IoT Integration: Adding Bluetooth or Wi-Fi for cloud data storage and remote access.
- Advanced Analysis: Implementing AI-based stress analysis and sophisticated signal processing.
- Wearable Design: Developing the device into a smart band or wearable form factor.

10. REFERENCES

1. Design And Implementation Of Portable ECG Monitoring (IJCRT).
2. Stress and Heart Rate Variability: A Meta-Analysis and Review (PMC).
3. GSM Based Real Time Human Health Monitoring and Alert System (IEEE).
4. Stress Analysis Based on Simultaneous Heart Rate Variability and EEG (PMC).
5. Arduino Official Website and AD8232 Datasheet.